Market Sentiment Classification of Financial Tweets

Text Mining — Spring Semester 2025/2026 · NOVA IMS

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1 Data Exploration

The training corpus contains 9,543 financial tweets, each labelled **Bearish (0)**, **Bullish (1)** or **Neutral (2)**. The corpus is clean: it has no missing values, no empty strings and no duplicate tweets, so no rows had to be removed and duplicates cannot leak across data splits. The distributed test corpus contains 2,388 unlabelled tweets with an explicit `id` column, which is used directly as the identifier in `pred_12.csv`. The handout mentions 299 test rows, but the distributed file contains 2,388 and predictions cover all of them.

1.1 Class distribution

The three classes are strongly imbalanced, as shown in Figure 1: Neutral accounts for 64.7% of the tweets (6,178), Bullish for 20.2% (1,923) and Bearish for 15.1% (1,442). This imbalance shapes the rest of the project. A classifier that always predicted Neutral would already be correct on almost two thirds of the corpus while never identifying a single Bearish or Bullish tweet, which are precisely the classes a market participant acts on. Overall accuracy is therefore not a sufficient measure of quality on this corpus, and the evaluation must weight performance on the minority classes explicitly. In practical terms, all data splits and cross-validation are stratified so that the minority classes remain represented, classifiers that support it receive balanced class weights, and per-class results are inspected for every model.

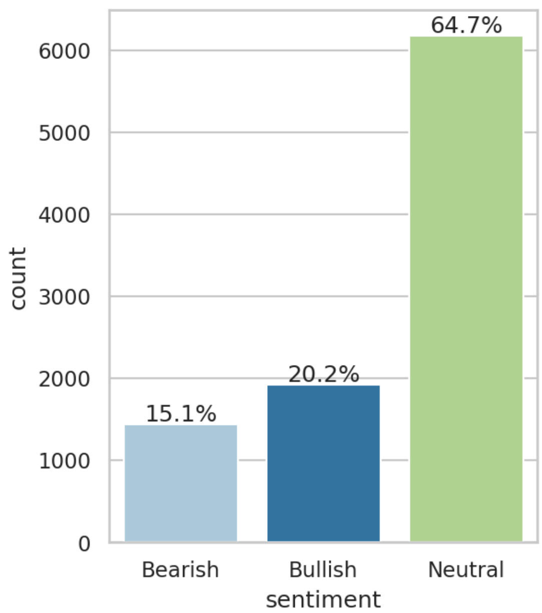

Figure 1: Distribution of sentiment classes in the training corpus.

1.2 Class vocabulary

Nineteen words, among them *stock(s)*, *market(s)*, *economy*, *china*, *trade*, *oil*, *price* and *earnings*, rank among the 50 most frequent words of all three classes simultaneously. They are general financial vocabulary rather than sentiment carriers, and they were excluded from the visualisations so that class-specific vocabulary becomes visible. The exclusion applies only to the figures; models receive the full text.

As shown in Figure 2, the remaining vocabulary separates the classes clearly. Bearish tweets are dominated by miss and downgrade language (*misses*, *cut*, *lower*, *falls*, *downgraded*) together with crisis terms such as *coronavirus*; Bullish tweets mirror them with beat and upgrade language (*beats*, *raised*, *higher*, *buy*, *target*, *eps*); Neutral tweets use the factual vocabulary of the corporate calendar (*results*, *reports*, *declares*, *dividend*, *conference*, *transcript*). The bigrams in the same figure sharpen this picture: after removing eleven boilerplate pairs common to all classes (*stock market*, *wall street*, *seeking alpha*), the leading Bearish and Bullish bigrams turn out to be built on identical stems that differ only in the verb, such as *target cut* against *target raised* and *misses revenue* against *beats revenue*. Sentiment in this corpus is thus encoded in short word pairs, which motivates including n-grams in the Bag-of-Words feature space.

Two further consequences follow. The strong lexical separability suggests that a simple word-frequency model will already be competitive, giving the more expensive contextual models a concrete baseline to beat. And because company and data-vendor names (*marketscreener*) rank among the most frequent tokens, a classifier could end up learning company identities rather than sentiment; cashtags are therefore normalised to a generic `ticker` token during preprocessing.

Figure 2: Word clouds (left) and bigram clouds (right) for Bearish, Bullish and Neutral tweets, after excluding the vocabulary shared by all three classes.

1.3 Financial-domain signals

Finance-specific tokens that generic text-cleaning pipelines delete carry class signal in this corpus, as shown in Figure 3. Percentage expressions occur in 21.2% of Bullish and 16.3% of Bearish tweets but in only 3.1% of Neutral ones: directional moves such as "+5%" express an opinion about a stock, while neutral reporting rarely quantifies one. Cashtags lean Bullish (25.1%, against roughly 13% in the other classes), as do bare numbers (47.7% against 29.8% in Neutral). Neutral tweets are the most news-like, with the highest incidence of URLs (49.8%), hashtags (11.0%) and `@mentions` (3.8%).

A cleaning step that deletes every digit, `%` and `$` would therefore erase the strongest signal separating opinionated from neutral tweets. For this reason the preprocessing stage normalises tickers, percentages and numbers to placeholder tokens instead of deleting them, and keeps negation words that standard stop-word lists would remove.

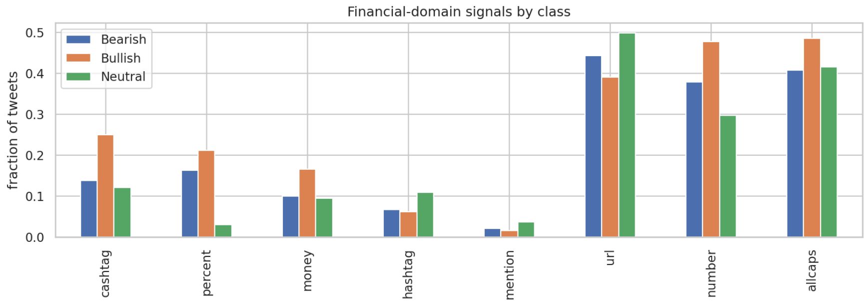

Figure 3: Fraction of tweets containing each finance-specific signal, by class.

1.4 Tweet length

Tweets are short and uniform: the median tweet has 11 whitespace tokens (mean 12.2, maximum 32), and per-class medians vary only between 11 and 12, so length itself does not separate the classes. The tail of the distribution sets the sequence-model configuration: the 95th percentile is 21 tokens and the 99th is 23, so a maximum sequence length of 32 word-tokens covers every tweet without truncation, and transformer tokenizers receive headroom above this value to absorb subword expansion.

2 Data Preprocessing

2.1 Corpus split

The labelled corpus was split into training, validation and internal test sets in a 70/15/15 ratio (6,679 / 1,432 / 1,432 tweets), stratified by class so that each subset preserves the 64.7 / 20.2 / 15.1 distribution, with a fixed random seed for reproducibility. An internal test set is needed because the distributed `test.csv` is unlabelled and serves only to produce the submission file: without a held-out portion of the labelled data, the validation set would be used both to select models and to estimate final performance, and that estimate would be optimistically biased. The protocol is therefore: candidate models are compared and tuned on the validation set, the selected final model is scored once on the internal test set, and the submission model is retrained on all 9,543 labelled tweets before predicting `test.csv`.

The integrity checks (duplicate, null and empty-text removal) run before the split, because a duplicate tweet present in two subsets would let the model see part of its evaluation data during training; on this corpus the checks removed nothing, so the guarantee is structural. Every step that learns parameters from data — vectorizers, vocabularies, scalers and the classifiers themselves — is fit on the training subset only, inside scikit-learn pipelines, so the same leak-free behaviour holds during cross-validation.

2.2 Cleaning pipeline

Six preprocessing techniques were implemented and demonstrated individually in the tests notebook: regular-expression normalisation (URLs, mentions, cashtags, percentages and numbers), lowercasing, special-character removal, stop-word removal, lemmatisation and stemming. They compose into a single finance-aware cleaner whose design follows from the exploratory findings of the previous section rather than from generic defaults:

- **URLs and `@mentions` are removed.** Twitter-shortened links are opaque identifiers and handles are names; neither contributes sentiment vocabulary.
- **Financial tokens are normalised, not deleted.** Cashtags become `ticker`, percentage expressions become `pct` and bare numbers become `num`. This keeps the strongest opinionated-versus-neutral signal found in the data (percentages appear in 21.2% of Bullish and 16.3% of Bearish tweets against

3.1% of Neutral) while collapsing thousands of distinct strings into three shared tokens. Replacing company tickers with a generic token also prevents a classifier from memorising company identities instead of sentiment vocabulary.

- **Lowercasing and special-character removal** run after the normalisation, so the inserted placeholder tokens survive the letters-only step.
- **Stop words are removed with negation kept.** The standard NLTK stoplist contains *not*, *no* and *never*; deleting them can invert the meaning of a tweet ("not bullish"), so a negation list is subtracted from the stoplist before filtering.
- **Lemmatisation is preferred over stemming.** Both were implemented and compared on samples; lemmatisation maps inflections to dictionary forms (*shares* ⟶ *share*) while stemming truncates to non-words (*raised* ⟶ *rais*). Since later sections inspect the most discriminative features, readable dictionary forms were retained.

As an example, the tweet "*$ORLY strong up 11.7% into large volume pocket w/ room to $360*" becomes "*ticker strong pct large volume pocket w room num*".

The cleaned text feeds the Bag-of-Words and word2vec representations only. Transformer encoders receive the raw text, because their subword tokenizers handle casing, punctuation and unknown words natively, and removing that surface information would discard cues those models exploit.